Solving Trig Equations

Exam code: 0607

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Solving trig equations

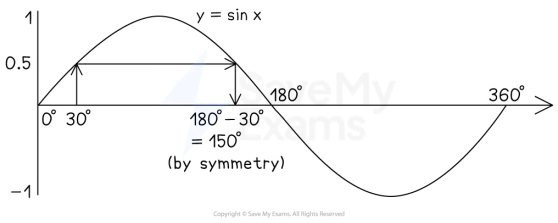
What are trig equations?

- **Trig equations** are equations involving $\sin x$, $\cos x$ and $\tan x$
- They often have **multiple solutions**
  - A **calculator** gives the **first solution**
  - You need to use **trig graphs** to find the others
  - The solutions must lie in the **interval** (range) of x given in the question, e.g. $0° \leq x \leq 360°$

How do I solve sin x = ...?

- Find the **first solution** of the equation by taking the **inverse sin function** on your **calculator** (or using an exact trig value)
  - E.g. For the first solution of the equation $\sin x = 0.5$ for $0° \leq x \leq 360°$
    - This gives $x = \sin^{-1}(0.5) = 30°$

- Then **sketch** the **sine graph** for the **given interval**
  - Identify the **first solution** on the graph
  - Use the **symmetry** of the graph to find **additional solutions**
  - E.g. For the equation $\sin x = 0.5$ for $0° \leq x \leq 360°$
    - Sketch the graph $y = \sin x$ for $0° \leq x \leq 360°$
    - Draw on $\sin(30) = 0.5$
    - By the symmetry, the new value of x is $180° - 30° = 150°$
    - The solutions are $30°$ or $150°$



- **Check** the solutions
  - E.g. For the equation $\sin x = 0.5$ for $0° \leq x \leq 360°$
    - Substitute $x = 30°$ and $x = 150°$ in to the calculator

Stuck on something?

Ask questions and get help with exactly what you're revising.

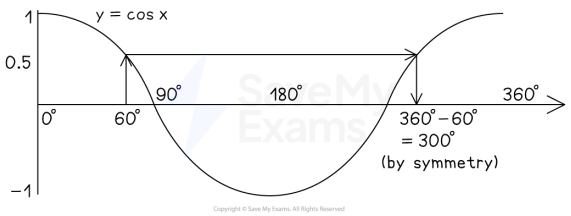
Got it

- Then $180 - x$ is an **obtuse** solution to the same equation

How do I solve cos x = ...?

- Find the **first solution** of the equation by taking the **inverse cos function** (or using an exact trig value)
  - E.g. For the first solution of the equation $\cos x = 0.5$ for $0° \leq x \leq 360°$
    - This gives $x = \cos^{-1}(0.5) = 60°$

- Then **sketch** the **cosine graph** for the **given interval**
  - Identify the **first solution** on the graph
  - Use the **symmetry** of the graph to find **additional solutions**
  - E.g. For the equation $\cos x = 0.5$ for $0° \leq x \leq 360°$
    - Sketch the graph $y = \cos x$ for $0° \leq x \leq 360°$
    - By the symmetry, the new value of x is $360° - 60° = 300°$
    - The solutions are $60°$ or $300°$



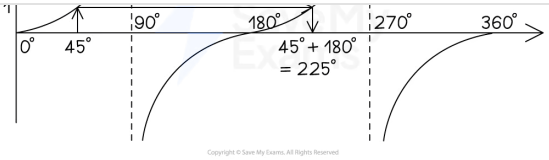
- **Check** the solutions
  - E.g. For the equation $\cos x = 0.5$ for $0° \leq x \leq 360°$
    - Substitute $x = 60°$ and $x = 300°$ in to the calculator
    - $\cos(60)$ and $\cos(300)$ both give a value of 0.5 so are correct

- In general, if x is a **solution** to $\cos x = ...$
  - Then $360 - x$ is **another solution** to the same equation

How do I solve tan x = ...?

- Find the **first solution** of the equation by taking the **inverse tan function** (or using an exact trig value)
  - E.g. For the first solution of the equation $\tan x = 1$ for $0° \leq x \leq 360°$
    - This gives $x = \tan^{-1}(1) = 45°$

- Then **sketch** the **tangent graph** for the **given interval**
  - Identify the **first solution** on the graph
  - Use the **periodic nature** of the graph to find **additional solutions**
  - E.g. For the equation $\tan x = 1$ for $0° \leq x \leq 360°$
    - Sketch the graph $y = \tan x$ for $0° \leq x \leq 360°$
    - By the periodic nature, the new value of x is $45° + 180° = 225°$



- **Check** the solutions
  - E.g. For the equation $\tan x = 0.5$ for $0° \leq x \leq 360°$
    - Substitute $x = 45°$ and $x = 225°$ in to the calculator
    - $\tan(45)$ and $\tan(225)$ both give a value of 1 so are correct
- In general, if x is a solution to $\tan x = ...$
  - Then $x + 180$ is **another solution** to the same equation

How do I rearrange trig equations?

- Trig equations may be given in a **different form**
  - Equations may require **rearranging** first
    - E.g. $2 \sin x - 1 = 0$ can be rearranged to $\sin x = \frac{1}{2}$
  - They can then be solved as usual

What do I do if the first solution from my calculator is negative?

- Sometimes the first solution given by the calculator for x will be **negative**
  - Continue sketching the graph to the **left** of the x-axis to help
  - Then find **solutions** that lie in the **interval** given in the question

Examiner Tips and Tricks

Make sure you know how to use the **inverse functions** (e.g. $\sin^{-1}$) on your calculator.

Worked Example

Use the graph of $y = \sin x$ to solve the equation $\sin x = 0.25$ for $0° \leq x \leq 360°$.

Give your answers correct to 1 decimal place.

Answer:

Use a calculator to find the first solution
Take the inverse sin of both sides

Mark on (roughly) where $x = 14.48$ and $y = 0.25$ would be

Draw a vertical line up to the curve
Draw another line horizontally across to the next point on the curve
Bring a line vertically back down to the x-axis

Find this value using the symmetry of the curve
Subtract 14.48 from 180

$$180 - 14.48 = 165.52$$

Give both answers correct to 1 decimal place

$$x = 14.5° \text{ or } x = 165.5°$$

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